

Ordered Tree

- A rooted tree in which children of each vertex are assigned an order.

Multway Trees Unit IV. 1 Dr. Rabins Porwal

1

Multway Trees

- Definition:** A general tree in which each node contains one or more keys/ values.
- B-tree:** A balanced, order m , multiway search tree in which each **non-root node** contains at least $(m-1)/2$ key values.
- 2-3 Tree:** The most elementary B-tree with 2 or 3 children per node.

Multway Trees Unit IV. 2 Dr. Rabins Porwal

2

2-3 Tree

- A general search tree which satisfies the following criteria:
 - Each **non-leaf node** in a 2-3 tree **must** have exactly 2 or 3 non-empty children, each of which is a 2-3 tree in turn.
 - All leaf nodes must be at the same level.
 - The nodes which have 2 children contain **exactly one** key value ' k '. Key values of all nodes in the left child of these nodes are less than ' k ' and the key values of all nodes in the right child of these nodes are greater than ' k '.
 - Nodes having 3 children contain 2 key values ' k ' and ' k_1 ' ($k < k_1$). The children are ordered – left, middle & right. Key values of all nodes in the middle subtree are more than k but less than k_1 , and the key values of all nodes in the right subtree are more than k_1 .

Multway Trees Unit IV. 3 Dr. Rabins Porwal

3

2-3 Tree (Contd...)

Example:

Multway Trees Unit IV. 4 Dr. Rabins Porwal

4

Note: If all non-leaf nodes in a 2-3 tree of height ' h ' have three children then the maximum number of elements (key values) would be $(3^{h+1} - 1)$ and number of nodes would be $(3^{h+1} - 1)/2$. Similarly, if each node of 2-3 tree has two children, then the tree has $(2^{h+1} - 1)$ elements at maximum. Therefore, a 2-3 tree with n elements having height ' h ' always satisfies the following condition:

$$(2^{h+1} - 1) \leq n \leq (3^{h+1} - 1)$$

Multway Trees Unit IV. 5 Dr. Rabins Porwal

5

Multway Search Trees

- Binary Search trees can be generalized directly to m -way search tree in which, for some integer m , called the **order of the tree**, each node has **at most** m children. If $k \leq m$, is the number of children, then nodes contain exactly $k - 1$ keys, which partition all the keys into key subsets. If some of these subsets are empty, then the corresponding children in the tree are empty.

Multway Trees Unit IV. 6 Dr. Rabins Porwal

6



Balanced Multiway Search Trees



- **Goal:** To devise m -way search tree that minimizes file access, *i.e.*, it is required to make the height of the tree as small as possible.
- This can be accomplished by ensuring:
 - No empty subtrees appear above the leaves (so division of keys into subsets is as efficient as possible)
 - All leaves should be at the same level (which guarantees the searching to terminate with about the same number of accesses)
 - Every node (except the leaves) has at least some minimal number of children. This requires that each node (except the leaves) has at least as many children as maximum possible.

Multiway Trees

Unit IV : 7

Dr. Rabins Porwal

7



B – Trees



- **Definition:** B-tree of order ' m ' is an m -way search tree in which
 - All leaves are at the same level.
 - All internal nodes except the root have **at most** m (non-empty) children & **at least** $m/2$ non-empty children.
 - The number of keys in each internal node is one less than the number of its children and these keys partition the keys in the children in the fashion of a search tree.
 - The root has at most m children but may have **as few as 2** if it is not a leaf, or none if the tree consists of the root alone.

Multiway Trees

Unit IV: 8

Dr. Rabins Porwal

8



Insertion in a B-Tree



- The B-trees are not allowed to grow at their leaves; instead, they are forced to grow at the root. The general method of insertion is as follows:
 - First a search is made to see if the new key is in the tree. This search will terminate in failure at a leaf.
 - The new key is added to the leaf node. If the node was not full, then insertion is finished.
 - When a key is added to a full node, then the node splits into two nodes on the same level except that the **median key** is not put into either of the new nodes but instead sent upward the tree to be inserted into the parent node.
 - When a search is later made through a tree, a comparison with the median key will serve to direct the search into proper subtree.
 - When a key is added to full root, then the root splits into two and the median key is sent upward becomes a new root.

Multiway Trees

Unit IV: 9

Dr. Rabins Porwal

9



Insertion in a B-Tree



- **Example:**
 - Construct a B-tree of order 5 with the following keys
a g f b k d h m j e s i r x c l n t u p
- **Solution:** On board

Multiway Trees

Unit IV: 10

Dr. Rabins Porwal

10



Construction of B-Tree



- Construct a B-tree of order 5 with the following keys:
a g f b k d h m j e s i r x c l n t u p
- Insertion of a, g

a	g		
---	---	--	--
- Insertion of f

a	f	g	
---	---	---	--
- Insertion of b

a	b	f	g
---	---	---	---

Multiway Trees

Unit IV: 11

Dr. Rabins Porwal

11



Construction of B-Tree (Contd...)



- Insertion of k

f			
a	b		
		g	k
- Insertion of d, h, m

f			
a	b	d	
		g	h k m

Multiway Trees

Unit IV: 12

Dr. Rabins Porwal

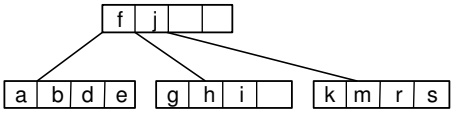
12



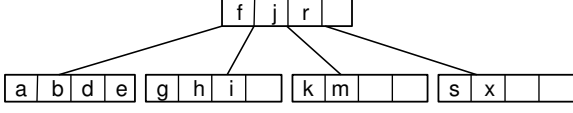
Construction of B-Tree (Contd...)



- Insertion of j, e, s, i, r



- Insertion of x



Multway Trees

Unit IV. 13

Dr. Rabins Porwal

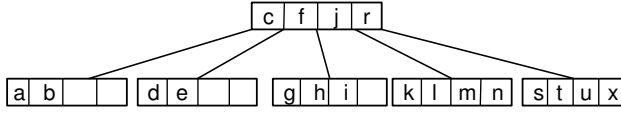
13



Construction of B-Tree (Contd...)



- Insertion of c, l, n, t, u



Multway Trees

Unit IV. 14

Dr. Rabins Porwal

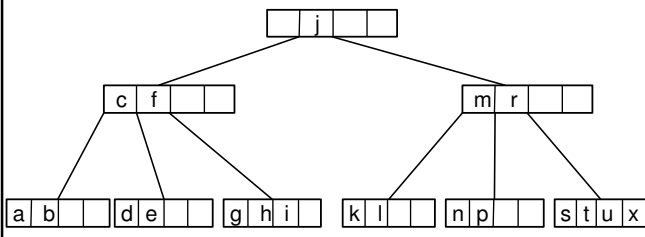
14



Construction of B-Tree (Contd...)



- Insertion of p



Multway Trees

Unit IV. 15

Dr. Rabins Porwal

15



Deletion in B-Tree



- If the key that is to be deleted is not the leaf then its *immediate predecessor or successor* is promoted into the position of the deleted key. The natural order of keys are guaranteed to be in the leaf nodes.
- If the leaf contains more than minimum number of keys, then it can be deleted with no further action.
- If the leaf contains the minimum number, then we first look at the two leaves (or in case of the node on the outside, one leaf) that are immediately adjacent and children of same node. If one of these has more than the minimum number of keys, then one of them can be moved into the parent node and the key from the parent is moved into the leaf where the deletion is occurring.

Multway Trees

Unit IV. 16

Dr. Rabins Porwal

16



Deletion in B-Tree (Contd...)



- If finally the adjacent leaf has only the minimum number of keys, then the two leaves and the median key from the parent can all be combined as one new leaf which will contain no more than the maximum number of keys allowed. If this step leaves parent node with too few keys, then the process propagates upwards. In this case, the last key is removed from the root, and then the height of the tree decreases.

Multway Trees

Unit IV. 17

Dr. Rabins Porwal

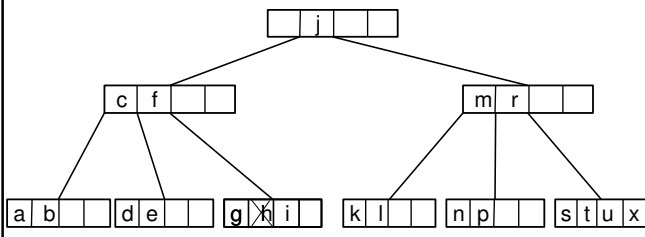
17



Deletion in B-Tree (Contd...)



- Deletion of h



Multway Trees

Unit IV. 18

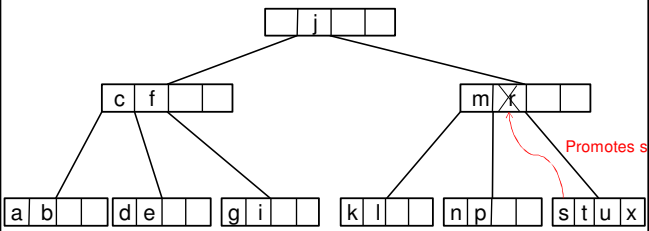
Dr. Rabins Porwal

18



Deletion in B-Tree (Contd...)

■ Deletion of r



Promotes s

Multway Trees

Unit IV. 19

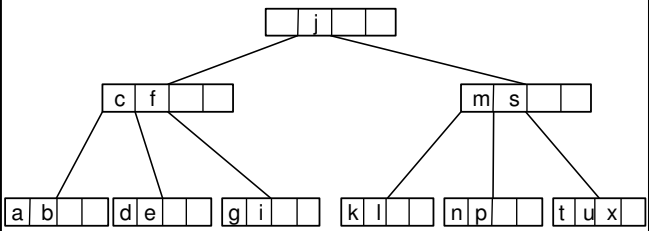
Dr. Rabins Porwal

19



Deletion in B-Tree (Contd...)

■ Deletion of r



Multway Trees

Unit IV. 20

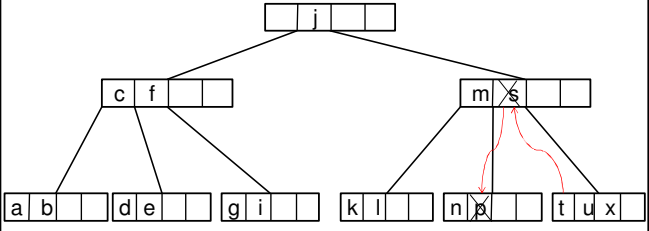
Dr. Rabins Porwal

20



Deletion in B-Tree (Contd...)

■ Deletion of p



Pull s down
Pull t up

Multway Trees

Unit IV. 21

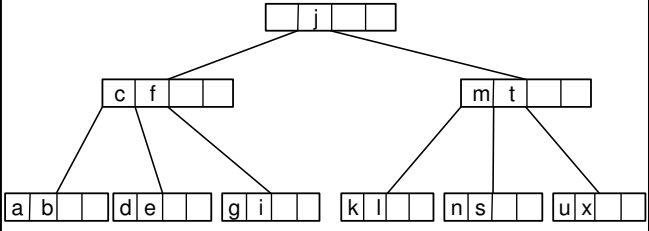
Dr. Rabins Porwal

21



Deletion in B-Tree (Contd...)

■ Deletion of p



Multway Trees

Unit IV. 22

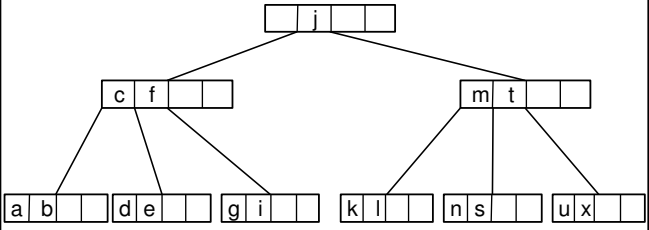
Dr. Rabins Porwal

22



Deletion in B-Tree (Contd...)

■ Deletion of d



Multway Trees

Unit IV. 23

Dr. Rabins Porwal

23



Deletion in B-Tree (Contd...)

■ Demonstrate the insertion of the key (1, 2, 3, 4, 8, 7, 6, 5, 9, 10, 11, 12, 16, 15, 14, 13) in a B-tree of order 5.

Multway Trees

Unit IV. 24

Dr. Rabins Porwal

24